



AI-Powered Diet-Based Disease Risk Prediction System for Nutrition-Related Diseases

BSc. in Software Engineering

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List of Acronyms/Abbreviations

Acronym	Meaning
AI	Artificial Intelligence
ALU	African Leadership University
API	Application Programming Interface
AUROC	Area Under the Receiver Operating Characteristic Curve
BMI	Body Mass Index
BSc.	Bachelor of Science
CFSVA	Comprehensive Food Security and Vulnerability Analysis
CGM	Continuous Glucose Monitor
CHW	Community Health Worker
CRUD	Create, Read, Update, Delete
DHS	Demographic and Health Survey
ERD	Entity Relationship Diagram
FAO	Food and Agriculture Organization
FCD	Food Composition Database
JWT	JSON Web Token
LMIC	Low- and Middle-Income Country
ML	Machine Learning
NCD	Non-Communicable Disease
NISR	National Institute of Statistics of Rwanda
ORM	Object-Relational Mapping
RDHS	Rwanda Demographic and Health Survey

Acronym	Meaning
REST	Representational State Transfer
SHAP	Shapley Additive Explanations
SMOTE	Synthetic Minority Oversampling Technique
SUS	System Usability Scale
UML	Unified Modeling Language
WHO	World Health Organization
XGBoost	Extreme Gradient Boosting

CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Diseases related to nutrition are a serious consideration in the public health arena in LMICs. The Rwanda Demographic and Health Survey (RDHS) 2025 reports that 37% of children aged 6 to 59 months are anemic and 13% of women of reproductive age are anemic (NISR, MoH & ICF, 2026). At the same time, 26% of women aged 15-49 are overweight or obese (up from 16.5% in 2010) and the estimated prevalence of adult diabetes in Rwanda is 5.5% (WHO, 2024). The MoH Rwanda allocated 358.15 billion Rwandan Francs for five years to NCD prevention and control, accounting for around 44% of the national death toll from NCDs (MoH Rwanda, 2020). It is this aspect of Rwanda's nutritional epidemiology that makes undernutrition an unsolvable problem, and that overnutrition is emerging on top of it that is unique. It is this that makes undernutrition an unsolvable problem in Rwanda, and it is the emergence of overnutrition on top of undernutrition that is unique, as noted by WHO and FAO (WHO/FAO, 2003).

The dual burden of malnutrition affliction is a challenge that African Leadership University (ALU) students and their community in Rwanda are confronted with and other institutions around the world. Diet is the greatest modifiable upstream risk for nutrition-related disease; however, people who want to know how their diet affects them have to sift through the disorganized bodies of information on diet: general recommendations, costly commercial precision-nutrition platforms tailored for Westerners, and reactive clinical screening that reveals a problem when it has already occurred.

The conventional interventions that have been used to tackle nutrition-related diseases include universal supplementation, fortified blended foods, counseling from community health workers, and screening at health clinics (UNICEF Rwanda, 2020). These approaches are important, but only detect risk when a person already has a measurable disease. In recent years, software-based solutions have arisen. In the UK, commercial precision-nutrition platforms like ZOE and in Israel DayTwo, apply machine learning algorithms, trained on CGM data, whole-genome sequencing, and microbiome sequencing to estimate individual disease risk (Bermingham et al., 2024; Zeevi et al., 2015). Dietary tracking apps like MyFitnessPal and Yazio let you track your diet and include macro summaries. No solution, however, has been found that seamlessly connects dietary intake, local food composition data

for Rwandan staples with multi-disease risk prediction in a one-stop, easy-to-use, cost-effective platform for African higher education and community settings.

New techniques like gradient-boosted decision trees and artificial intelligence that can explain their results have opened up new opportunities to make predictions in the field. Using publicly-available survey data, machine learning models can predict the risk of inadequate micronutrient intake in LMICs without the need for costly biomarkers, as shown by Voukelatou et al., 2026. Theodore Armand et al., (2024) examined 1,073 papers and found that XGBoost and other ensemble models outperform the state-of-the-art on tabular health-oriented data, are interpretable and have computational costs low enough to fit the resources of free cloud computing important traits for African institutions that are often resource constrained.

1.2 Problem Statement

Despite the availability of various software tools for nutrition tracking and disease prediction, individuals in Rwanda lack access to a system that predicts their personal risk of nutrition-related diseases from what they actually eat, using foods that exist in the local food system, at a cost they can afford. The current solutions either need to be costly and are not yet in place at population scale, or completely ignore the relation between diet and disease.

Two closest existing solutions and their limitations:

1. Commercial Precision Nutrition Platforms (ZOE, DayTwo): These platforms leverage machine learning algorithms trained on extensive biological information, such as continuous glucose monitor data, gut microbiome samples, blood lipid panels, and genetic testing, to forecast individual dietary reactions and health outcomes. These platforms utilize AI algorithms which are trained using vast information on biological data, including continuous glucose monitor readings, microbiome sequencing, blood lipid panels, and genetic profiling, to anticipate personalized dietary responses and disease risk. They take up price points ranging from USD 300 to USD 2,000 per user when equipment, sequencing and subscriptions are considered, and are only trained on Western foods, which are not relevant to the Rwandan diet (particularly ubugali, isombe, beans, and cassava), and focus only on one disease (mainly type 2 diabetes) and do not

even consider the undernutrition side of the Rwandan burden at all. They target precision but do not take into account those most affected by malnutrition in Rwanda.

2. LMIC Micronutrient Prediction (Voukelatou et al., 2026): Voukelatou et al. (2026) they have shown that machine learning models, based on household survey information, socio-economic indicators, and climate data, can accurately forecast micronutrient intake inadequacy (i.e., zinc, folate, iron, vitamin A, B12) in Ethiopia and Nigeria. This is the most LMIC-related work available which is closest to the current project. It still focuses on intake adequacy instead of disease outcomes, is household and regional-level, only covers the undernutrition side of the burden, doesn't have a recommendation engine, and doesn't have any coverage of Rwanda.

The Gap: There is no current system that combines (1) individual level dietary intake logging adapted to the local food landscape (Rwanda), (2) machine learning-based prediction of risk on multiple nutrition-related diseases ranging from undernutrition to overnutrition, (3) locally appropriate dietary recommendations, respecting local food availability and affordability, (4) accessible web-based delivery, with an accessible cost-effective platform suitable for resource-constrained African contexts. To fill this gap, this research aims to build an integrated web-application based on a Rwandan Food Composition Database and a fine-tuned XGBoost multi-output model for intelligent disease risk prediction based on dietary patterns.

1.3 Project's Main Objective

To develop a web-based AI-powered diet-based disease risk prediction system that integrates dietary intake logging with a Rwandan Food Composition Database, multi-output machine learning prediction of risk for anemia, type 2 diabetes, and overweight, and locally appropriate dietary recommendations.

1.3.1 Specific Objectives

To analyze current dietary and nutritional risk patterns at ALU and the surrounding community through stakeholder interviews (5 nutritionists or health workers) and student surveys (30 students).

To develop an integrated web platform with dietary intake logging, food-to-nutrient conversion using a Rwandan Food Composition Database of at least 100 foods, multi-output

XGBoost prediction, and locally appropriate dietary recommendations, using free-tier infrastructure (Theodore Armand et al., 2024).

To evaluate system performance through technical metrics (AUROC ≥ 0.75 per disease, response time < 3 seconds), usability testing (SUS score ≥ 70), and user acceptance testing with 15–20 participants.

1.4 Research Questions

What are the primary dietary patterns and perceived nutrition-related risks among the ALU student population and surrounding community?

How effectively can a multi-output XGBoost model predict individual risk of anemia, type 2 diabetes, and overweight from dietary intake data using a Rwandan Food Composition Database and publicly available training data?

How does an integrated web platform with dietary logging, multi-disease risk prediction, and locally appropriate dietary recommendations improve users' understanding of their personal nutritional risk compared to generic dietary guidelines?

1.5 Project Scope

Geographical Scope: The platform will be developed and tested at African Leadership University, Rwanda campus. User acceptance testing will be conducted with a pilot group of 15–20 participants in Kigali, comprising students from various academic programs and 2–3 health workers or nutritionists. This sample represents the broader ALU community while remaining realistic for a 16-week development timeline.

The system encompasses dietary intake logging, food-to-nutrient conversion, multi-output disease risk prediction, and locally appropriate dietary recommendations via a web application with email notifications. The research explicitly excludes integration with national health information systems such as DHIS2, native mobile application development, real-time biomarker data integration (continuous glucose monitors, lab panels), genetic or microbiome-based prediction, and SMS notification infrastructure. The machine learning approach targets multi-output classification using XGBoost on tabular dietary and demographic features, balancing accuracy with computational efficiency for

resource-constrained deployment. Unlike commercial precision nutrition systems such as ZOE (Bermingham et al., 2024), this system focuses on accessible non-biomarker inputs (diet, demographics, basic anthropometrics) rather than continuous glucose monitor or microbiome data. Unlike micronutrient prediction models such as Voukelatou et al. (2026), this system predicts individual disease outcomes (anemia, diabetes risk, overweight) rather than household-level intake inadequacy.

1.6 Significance and Justification

This platform will allow ALU students and community to more easily access the information they need if they are at risk to nutrition-related diseases based on what they eat, removing the current fragmentation between general nutrition guidelines, costly commercial precision nutrition platforms, and reactive clinical screening and the multi-output machine learning model will provide a unified assessment of risk across both sides of the dual burden, with the locally appropriate recommendation engine ensuring that the recommendations consider foods available and affordable to the target population. For health workers and nutritionists, the system will also provide insights into dietary patterns that are linked to higher risk, allowing them to provide guidance before the onset of disease, and the documentation of dietary trends among users will lay the foundation for insights into the population level. More broadly, this study contributes a repeatable framework for integrating predictive nutrition tools in resource-limited environments in Africa, since the successful implementation of the free tier cloud services and open-source tools (XGBoost, scikit-learn, React, Flask, Render free tier) shows that sophisticated AI-based tools can be deployed without a large up-front investment, mitigating concerns raised by Voukelatou et al. (2026) regarding sustainability in LMIC applications.

1.7 Research Budget

The following table presents the estimated budget using free tiers where possible:

Item	Purpose	Cost (USD)
Google Colab	Free GPU for model training (XGBoost)	\$0
Hugging Face Hub	Pretrained tokenizer access (optional)	\$0

Item	Purpose	Cost (USD)
Render (Free Tier)	Web application hosting	\$0
PostgreSQL (Render)	Database hosting	\$0
GitHub	Version control and CI/CD	\$0
SendGrid (Free Tier)	Email notifications (100/day)	\$0
Figma (Free Tier)	UI/UX design	\$0
Google Forms	Survey data collection	\$0
DHS Program microdata	RDHS 2019–20 and 2025 datasets (request)	\$0
FAO INFOODS, Congo Basin FCT	Public food composition data	\$0
TOTAL		\$0

1.8 Research Timeline

The project follows a 16-week timeline:

Activity	Weeks 1-4	Weeks 5-10	Weeks 11-13	Weeks 14-16
Requirements gathering & stakeholder interviews	■			
Literature review & system design	■			
Rwandan Food Composition Database compilation (100 foods)	■	■		
RDHS microdata acquisition & exploratory analysis	■	■		
Dietary logging module (frontend + backend)		■		
Food-to-nutrient conversion service		■		

Activity	Weeks 1-4	Weeks 5-10	Weeks 11-13	Weeks 14-16
XGBoost multi-output model training (Google Colab)			■	
Recommendation engine development			■	
ML model integration into backend			■	
System integration & testing				■
User acceptance testing (15-20 users)				■
Documentation & final report				■

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction and Historical Background

This literature review aims to discuss various software systems that encompass dietary disease risk prediction, machine learning-based nutritional surveillance systems, and personalized recommendation systems for dietary interventions in higher education and LMIC settings. The review includes recent peer-reviewed publications (2020–2026) found through keyword search on Google Scholar, PubMed, and Nature Portfolio, with the following keywords: "diet disease prediction machine learning," "precision nutrition AI," "LMIC micronutrient prediction," "XGBoost nutrition," "dual burden of malnutrition," and "food composition Rwanda. The following eight core papers were selected for detailed analysis, due to their direct relevance with the research objectives: precision nutrition machine learning (Bermingham et al, 2024; Zeevi et al, 2015), LMIC micronutrient prediction (Voukelatou et al, 2026), Rwandan nutrition surveillance (NISR, MoH & ICF, 2026; MoH Rwanda, 2020), retrospective nutrition machine learning in Rwanda (Munyemana et al, 2024), systematic reviews of AI in nutrition (Theodore Armand et al, 2024), conceptual frameworks for precision nutrition (Rojanaphan, 2024).

The systems used for diet-disease prediction have progressed from simple manual nutrient evaluation to more and more advanced machine learning platforms. The field of personalized nutrition came to the fore in about 2015 with the work of Zeevi et al. (2015) at the Weizmann Institute showing that the composition of the gut microbiome can be used to predict an individual's postprandial glucose response, which has been translated into the commercialization of DayTwo. The COVID-19 pandemic has not only driven greater interest in remote, software-driven health solutions, but also highlighted major gaps in nutrition surveillance and prediction infrastructure in LMIC settings. Multi-disease prediction from dietary data has a shorter but quick history when it comes to machine learning techniques. In a randomized controlled trial, published in the journal *Nature Medicine* by Bermingham et al. 2024, it was confirmed that the ZOE METHOD program produced a personalized dietary score that was beneficial for cardiometabolic outcomes. To better understand the performance of gradient boosted classifiers on real health data, Theodore Armand et al. (2024) analysed 1,073 studies and found that these models, in particular XGBoost, deliver state-of-the-art performance on tabular health data and that they are easily interpretable and lightweight

enough to be deployed on free cloud resources, especially for African institutions that have constrained computational budgets.

2.2 Overview of Existing Systems

This section reviews software systems that represent the closest solutions to the proposed platform, examining where they fall short and identifying gaps to be addressed.

2.2.1 Commercial Precision Nutrition Platforms (ZOE, DayTwo)

ZOE METHOD was published in the randomized controlled trial by Bermingham et al. (2024) in the Journal of Nature Medicine in collaboration with King's College London, Massachusetts General Hospital, Stanford University, and the Harvard T. H. Chan School of Public Health. There were approximately 4 million blood glucose measurements, 56,000 triglyceride measurements, and 12 terabytes of microbiome data in the PREDICT trials for training a personalized dietary scores model. The trial showed beneficial effects on cardiometabolic parameters versus the typical U.S. Dietary Guidelines. Based on the work of Zeevi et al., 2015, DayTwo predicts glucose responses for each individual after meals, based on gut microbiome composition, and clinical trial validates predictions on type 2 diabetes management. These technologies are at the cutting edge of diet-disease prediction but all involve hardware (Continuous Glucose Monitor), biological sample analysis (microbiome sequencing) or laboratory measurement (lipid panels) that cost anywhere from USD 300 to USD 2,000 per user. In addition, they are only trained on food databases from the West, only focus on a single disease, in this case mainly type 2 diabetes and do not account for undernutrition at all.

2.2.2 LMIC Micronutrient Prediction (Voukelatou et al., 2026)

In the paper published in Nature Scientific Reports (2026), Voukelatou et al. designed a machine learning model that uses data from household surveys, socioeconomic indicators, and climate data to predict the risk of households being exposed to less than the recommended daily intake of five micronutrients (zinc, folate, iron, vitamin A, B12) at the household and first-level administrative unit level in Ethiopia and Nigeria. The work represents an important methodological cross-over from one country to another where data is available. This is the most recent published research project in the domain of machine learning and LMICs that is close to the present project. The system, however, does not go

beyond the point of intake adequacy and does not extend to the individual level, only the undernutrition side of the burden, user-friendly risk maps for policy use, and does not include the country of Rwanda.

2.2.3 Consumer Nutrition Tracking Apps (MyFitnessPal, Yazio)

The dominant players in the dietary logging market are consumer nutrition tracking applications and are used around the world. These platforms enable food logging, a searchable food database, a summary of macronutrients and calories from the food, tracking daily and weekly nutritional goals, and premium versions include image recognition for food identification. But they have some drawbacks for the situation of the proposed research. They were trained using western commercial food databases which lack the representation of African food, and give no disease risk prediction, no personalised nutrition advice, and are only affordable with a subscription fee of USD 4 – USD 20 per month for advanced features. Moreover, the composition information for their food includes no meaningful amount of information about Rwandan staples, which precludes them from being useful for nutrient profiling in the context of this document.

2.2.4 NHANES-based Disease Risk Prediction Models

The United States National Health and Nutrition Examination Survey (NHANES) database is employed in the creation of machine learning models to predict the outcomes of disease from dietary intake in an entire body of academic literature. In the Nutrition Journal, Chen et al. (2025) applied five survival ML models to predict all-cause mortality in NAFLD patients based on dietary intake variables, achieving the highest accuracy with Gradient Boosting Machine (GBM). The work shows that it is possible to achieve meaningful performance levels in terms of diet-based machine learning prediction of disease outcomes. However, the population (adults in the United States of America), the food composition basis (USDA database), and the focus of the disease (single-outcome mortality) restrict transferability. In addition to these models, there are published research artefacts or models that are not systems that can be deployed by individuals or health workers.

2.3 Summary of Reviewed Literature

Study	Focus Area	Key Finding	Relevance
Bermingham et al. (2024)	Precision nutrition RCT (ZOE METHOD)	Personalized ML scoring improves cardiometabolic outcomes	Validates ML approach to diet-disease prediction
Zeevi et al. (2015)	Personalized glucose prediction (DayTwo basis)	Individual glycemic responses are predictable from diet + microbiome	Establishes diet → disease ML feasibility
Voukelatou et al. (2026)	LMIC micronutrient ML	Survey + climate data predicts inadequate intake at regional level	Closest LMIC ML work; informs methodology
Munyemana et al. (2024)	Rwanda retrospective ML on child nutrition outcomes	ML models can analyze RDHS data effectively for nutrition prediction	Demonstrates ML feasibility on Rwandan survey data
Chen et al. (2025)	NHANES survival ML on diet-NAFLD mortality	XGBoost outperforms baselines on diet → outcome prediction	Justifies XGBoost selection
Theodore Armand et al. (2024)	Systematic review of AI in nutrition	Gradient-boosted models dominate tabular nutrition ML	Validates model selection
Rojanaphan (2024)	Precision nutrition framework (IJSRM)	Multi-modal ML can integrate diet + biomarkers	Conceptual scaffold; not LMIC-applicable
NISR, MoH & ICF (2026)	RDHS 2025 national survey	Current Rwanda dual-burden statistics	Defines target population and outcomes

2.4 Strengths and Weakness of the Existing System(s)

System	Strengths	Weaknesses
ZOE / DayTwo (Bermingham et al., 2024; Zeevi et al., 2015)	Largest deployed precision-nutrition datasets; RCT-validated; sophisticated ML	Requires CGM + microbiome / DNA testing (USD 300–2,000); Western foods only; single-disease focus

System	Strengths	Weaknesses
Voukelatou et al. (2026)	LMIC-focused; uses publicly available data; cross-country transferability	Stops at intake adequacy, not disease; household / regional only; undernutrition only; not Rwanda
MyFitnessPal / Yazio	User-friendly logging; large food database; image recognition	No disease prediction; Western foods only; no personalization to risk; subscription costs
NHANES-based ML (Chen et al., 2025)	Rigorous methodology; large dataset; multiple algorithm benchmarks	U.S. population only; USDA food basis; not deployed; single disease per study
Munyemana et al. (2024)	Rwanda-focused; uses RDHS data; demonstrates ML feasibility on local data	Retrospective only; targets a single childhood outcome; not deployed; no recommendation engine

2.5 General Comment and Conclusion

The literature search shows that there is a growing number of individual components of an integrated diet-based disease risk prediction platform available, but no platform that integrates all the required functions for the dual-burden setting in Rwanda. However, five main gaps emerge from the analysis: first, systems that are based on precision-nutrition are focused on a single disease, expensive biomarkers, whereas LMIC micronutrient models end at intake adequacy and are applied at the regional level; second, the African context gap: existing commercial precision-nutrition solutions assume Western dietary patterns and high income consumer infrastructure, while Theodore Armand et al. (2024) have noted that focused and cost-effective solutions to undernutrition are necessary in resource-constrained contexts without recurring subscription fees; third, the absence of dual burden coverage in any single system, as precision-nutrition solutions focus on NCDs, and LMIC studies focus on undernutrition; fourth, lack of awareness of food composition across all systems reviewed, where commercial solutions rely on USDA or European food composition data, and LMIC studies rely on food group proxies instead of actual nutrient values for Rwandan foods; and fifth, the cost-effectiveness gap, with precision-nutrition solutions costing USD 300 to USD 2,000. All five gaps are treated in this research by building an integrated web platform that leverages free tier infrastructure (XGBoost, Google Colab, Render) and a Rwandan Food

Composition Database built from the Food Composition tables from FAO INFOODS, Congo basin and Uganda, as well as a multi-output model trained using RDHS data to predict anemia, type 2 diabetes risk, and overweight.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the system analysis and design for the AI-Powered Diet-Based Disease Risk Prediction System. The design follows a pragmatist research philosophy, focusing on practical solutions to real-world problems. The development approach combines Design Science Research methodology (Vom Brocke et al., 2020) with Agile development practices, enabling iterative refinement based on user feedback within the 16-week timeline.

The system is designed as a three-tier web application with a React.js frontend, Flask REST API backend, and PostgreSQL database, integrated with a fine-tuned XGBoost multi-output model for diet-driven disease risk prediction. This architecture balances functionality with the resource constraints identified by Voukelatou et al. (2026) for LMIC implementations.

3.2 Research Design (including the development model used)

The development model follows Agile methodology with one-week sprints, incorporating the Design Science Research build-evaluate cycle advocated by Vom Brocke et al. (2020). The research approach is both deductive, in that it tests the hypothesis that a multi-output XGBoost model achieves an AUROC of at least 0.75 per disease based on benchmarks from Chen et al. (2025) and Theodore Armand et al. (2024), and inductive, in that it discovers user requirements and dietary patterns through stakeholder interviews and surveys.

3.2.1 Data Collection and Analysis

The system development requires three data collection phases to inform the platform design, the Rwandan Food Composition Database, and the machine learning model. Requirements data will be gathered through semi-structured interviews with 5 nutritionists or health workers (30–45 minutes each, covering common nutritional risks, gaps in existing guidance, and required features) and online Google Forms surveys of 30 ALU students (covering dietary habits, exposure to nutrition information, and preferred platform features); the qualitative and quantitative responses will inform functional requirements, interface design, and priority diseases for Phase 1 prediction. Food composition data will be compiled from publicly available sources including the FAO INFOODS West African Food Composition Table, the Congo Basin Food

Composition Table (CIFOR-ICRAF, 2023), the Ugandan Food Composition Table, and USDA FoodData Central, targeting at least 100 foods covering Rwandan staples such as ubugali, isombe, beans, sorghum, cassava, plantain, sweet potato, rice, and maize, with values for energy, macronutrients, and key micronutrients.

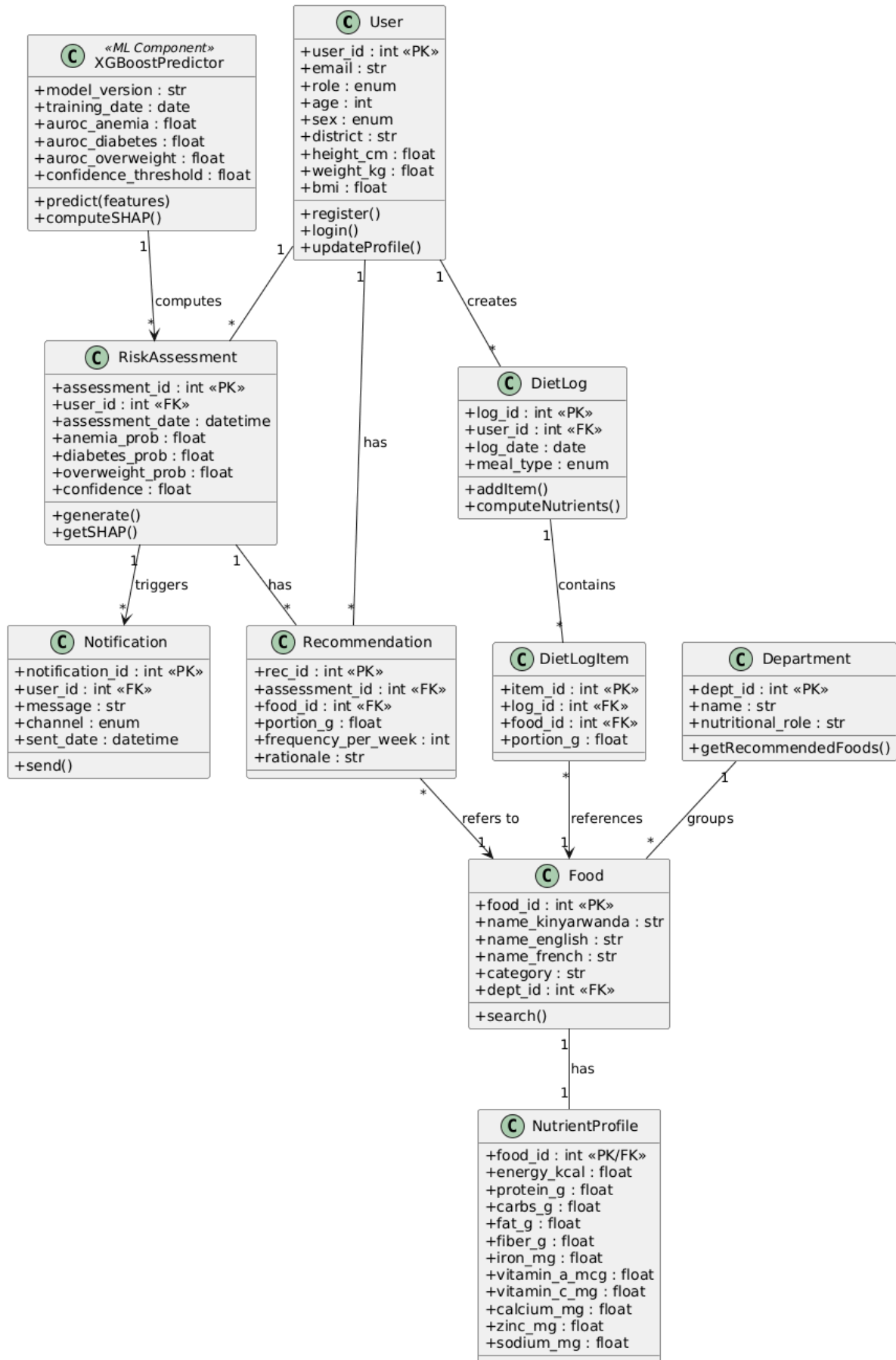
Training data for the XGBoost prediction model will be derived from the Rwanda Demographic and Health Survey 2019–20 and 2025 microdata, obtained via the DHS Program after a research access request and containing approximately 12,949 women-level records in 2019–20 with comparable sample in 2025, including dietary indicators, anthropometric measurements, anemia status from blood hemoglobin testing, and demographic context; the labeled dataset will be split 80/20 for training and validation, with class-weighted training applied to address the class imbalance arising from anemia, diabetes, and overweight as minority outcomes, and the compiled food composition data will support feature engineering to convert dietary indicators into estimated nutrient intakes. The trained model will be integrated into the Flask backend to predict risk for each user submission with inference under 500 milliseconds following Chen et al. (2025), and SHAP values will be computed alongside each prediction to identify the dietary features driving the risk score, providing the explainability layer recommended by Theodore Armand et al. (2024) for clinical and educational trust.

3.3 Class Diagram

The class diagram shows the main classes of the Diet-Based Disease Risk Prediction System, and the associations used to convert dietary intake into the individual disease risk and locally appropriate recommendations. They are fed into a prediction model and carry attributes of the User (students, community participants, administrators) including its role and permissions, and its demographic and anthropometric attributes (age, sex, district, height_cm, weight_kg, bmi). A User is connected to DietLog via a one to many relationship, and a DietLog is, in turn, composed of one to many DietLogItem entries, each of which is connected to the Food class with a Food intake portion_g value. The Food class stores names in Kinyarwanda, English, and French, the food category, and links with one-to-one relationship to the NutrientProfile class, which stores the energy, protein, carbohydrates, fat, fibre, iron, vitamin A, vitamin C, calcium, zinc, and sodium values per 100g of food.

The output side of the system focuses on RiskAssessment, a component designed for a User and calculated by the XGBoostPredictor component and providing three density probabilities (anemia_prob, diabetes_prob, overweight_prob) based on the User's recent DietLogs and demographic context. Each RiskAssessment is related in one-to-many to Recommendation entries, one Recommendation for each Food, with each referencing a specific amount of grams, a specific frequency, and a plain-language description of the rationale, that the system automatically generates a Notification to the user (via email) when an assessment activates. The XGBoostPredictor wraps up the machine learning model, and includes attributes for version, training date, and per-disease performance metrics (auroc_anemia, auroc_diabetes, auroc_overweight), and the model's predict() method returns the three probability scores, as well as SHAP values to explain the model's decisions; the FoodCategory class groups a food by its nutritional role and exposes getRecommendedFoods() to extract a food suitable for closing a nutrient gap. This architecture tackles the fragmentation of food-to-nutrient conversion and biomarker dependency problem statement by feeding a logged meal into a food-to-nutrient conversion model, followed by an XGBoost model that generates predictions, and finally, dietary suggestions that are locally relevant.

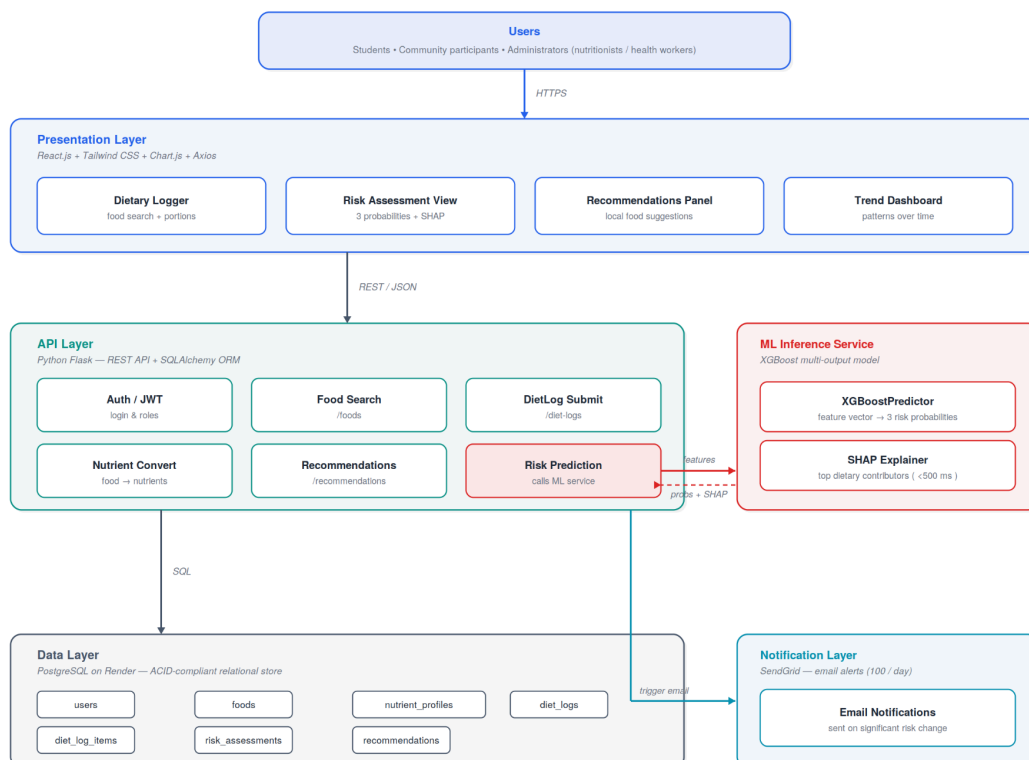
Figure 1: Class Diagram



3.4 System Architecture

The system follows a three-tier architecture. The Presentation Layer (React.js with Tailwind CSS) provides four interfaces: Dietary Logger (food search and portion entry), Risk Assessment View (displays the three risk probabilities with SHAP-derived feature contributions), Recommendations Panel (lists locally appropriate dietary suggestions), and Trend Dashboard (shows the user's dietary patterns and risk evolution over time). The API Layer (Python Flask) implements six REST endpoints handling authentication (JWT), food search, dietary log submission, food-to-nutrient conversion, risk prediction (with the XGBoost service), and recommendation generation. The Data Layer uses PostgreSQL on Render, storing users, dietary logs, foods, nutrient profiles, risk assessments, and recommendations. The Notification Layer employs SendGrid for email alerts when risk scores change significantly. The entire stack deploys on Render's free tier, achieving cost-effectiveness for resource-constrained institutions while maintaining ACID compliance.

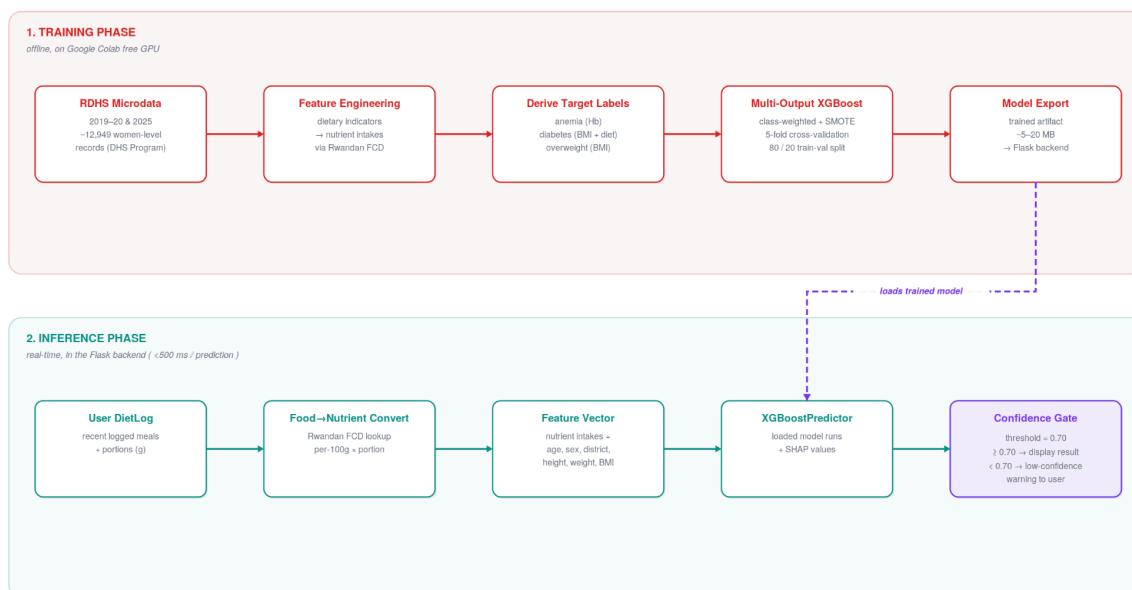
Figure 2: System Architecture Diagram



The machine learning architecture comprises two phases. The Training Phase loads RDHS 2019–20 and 2025 microdata, performs feature engineering to convert dietary indicators into

estimated nutrient intakes using the Rwandan Food Composition Database, derives target labels (anemia from hemoglobin levels, diabetes risk from BMI and dietary risk patterns, overweight from BMI categories), and trains a multi-output XGBoost model on Google Colab's free GPU. The fine-tuning process applies class-weighted training to address imbalance (since each outcome is a minority class), uses 5-fold cross-validation, and exports the trained model (approximately 5–20 MB) to the Flask backend. The Inference Phase processes incoming dietary logs in real-time: the user's recent DietLog is converted to a nutrient profile using the Rwandan FCD, combined with demographic and anthropometric features, and passed to the XGBoost model. The model outputs three probability scores along with SHAP values for the top-contributing features. Inference completes in under 500 milliseconds per prediction, enabling real-time feedback. A confidence threshold of 0.70 separates auto-displayed predictions from those flagged with a low-confidence warning to the user.

Figure 3: Model Architecture Diagram



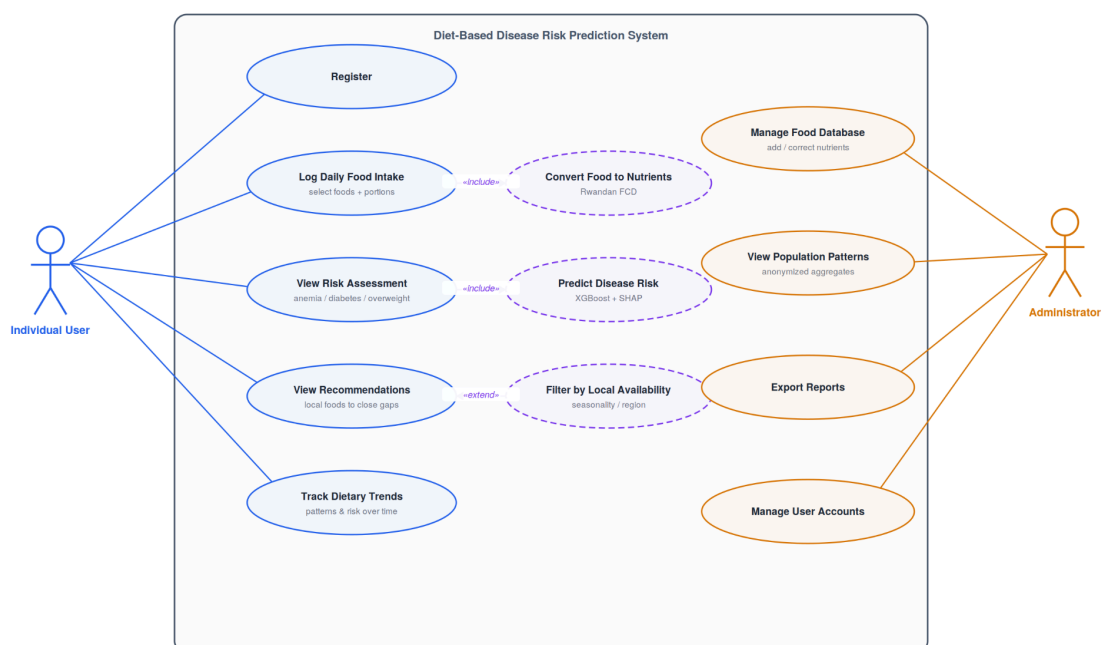
3.5 UML Diagrams

The use case diagram illustrates the functional interactions between two actor types and the Diet-Based Disease Risk Prediction System. Individual Users (blue) can register, log their daily food intake which includes selecting foods from the Rwandan FCD and specifying

portions, view their current risk assessment showing probabilities for anemia, type 2 diabetes, and overweight along with the dietary factors driving each score, view personalized recommendations of locally available foods that would close their identified nutrient gaps, and track their dietary trends and risk evolution over time. Administrators (orange) — typically nutritionists, health workers, or research staff — manage the food composition database by adding new foods or correcting nutrient values, view anonymized population-level dietary patterns to identify community-wide nutritional issues, export reports for institutional use, and manage user accounts.

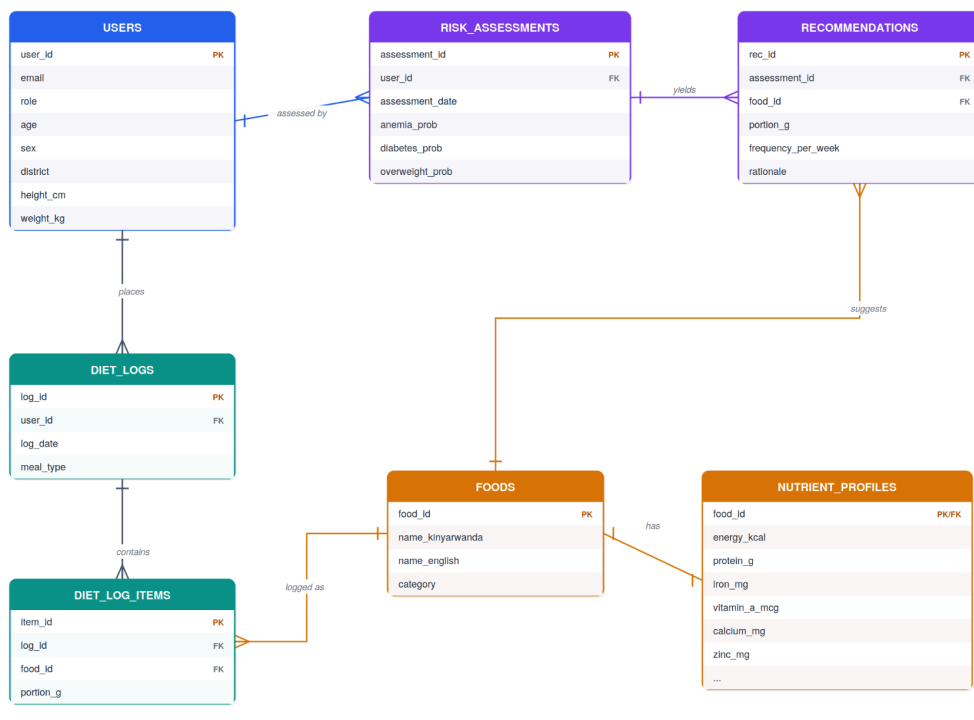
Three system-initiated use cases (dashed ovals) occur automatically. "Convert Food to Nutrients" is included whenever a user logs a meal, executing the food-to-nutrient conversion via the Rwandan FCD. "Predict Disease Risk" is included whenever a user views their risk assessment, executing the XGBoost model and producing the three probability scores with SHAP explanations. "Filter by Local Availability" extends the View Recommendations workflow, applying seasonality and regional availability constraints to the recommendation engine output. These automated workflows directly address the fragmentation, biomarker-dependency, and Western-food-bias problems identified in the problem statement through integrated, automated, locally appropriate intelligent processing.

Figure 4: Use Case Diagram



An Entity Relationship Diagram (ERD) for the database is also required and should show the following entities with their primary keys and key foreign-key relationships: USERS (user_id PK, email, role, age, sex, district, height_cm, weight_kg) — DIET_LOGS (log_id PK, user_id FK, log_date, meal_type) — DIET_LOG_ITEMS (item_id PK, log_id FK, food_id FK, portion_g) — FOODS (food_id PK, name_kinyarwanda, name_english, category) — NUTRIENT_PROFILES (food_id PK/FK, energy_kcal, protein_g, iron_mg, vitamin_a_mcg, etc.) — RISK_ASSESSMENTS (assessment_id PK, user_id FK, assessment_date, anemia_prob, diabetes_prob, overweight_prob) — RECOMMENDATIONS (rec_id PK, assessment_id FK, food_id FK, portion_g, frequency_per_week, rationale).

Figure 5: Entity Relationship Diagram (ERD)



3.6 Development Tools

Category	Tool	Purpose	Justification
Frontend	React.js	UI Development	Component-based architecture; large ecosystem
Frontend	Tailwind CSS	Styling	Utility-first; rapid prototyping

Category	Tool	Purpose	Justification
Frontend	Chart.js	Data Visualization	Risk dashboard and trend charts
Frontend	Axios	HTTP Client	Promise-based API calls
Backend	Python 3.10	Server Language	ML ecosystem compatibility
Backend	Flask	Web Framework	Lightweight; RESTful API support
Backend	SQLAlchemy	ORM	Database abstraction
Backend	JWT	Authentication	Stateless; scalable
ML	XGBoost	Multi-output classifier	Industry standard for tabular health data
ML	scikit-learn	Pipelines, metrics, SMOTE	Standard ML library
ML	SHAP	Explainability	Feature contribution per prediction
ML	Google Colab	GPU training	Free T4 GPU; sufficient for XGBoost training
Database	PostgreSQL	RDBMS	ACID compliance; free tier on Render
DevOps	Git / GitHub	Version control	Collaboration; CI/CD integration
DevOps	Render	Hosting	Free tier; PostgreSQL included
DevOps	SendGrid	Email	Free tier (100 emails/day)
Evaluation	Google Forms	Surveys	SUS questionnaire and feedback collection
Evaluation	Python / Pandas	Analysis	Statistical evaluation of results

3.7 Ethical Considerations

The project research involves human participants and the collection of dietary and basic anthropometric data, which requires careful attention to ethical aspects throughout the project lifecycle. All research activities will follow the institutional research ethics guidelines of ALU and will be approved in advance before data collection, with additional ethics clearance by the Rwanda National Ethics Committee, if it is determined that the research will fall within the definition of "health research" under the Rwandan laws. All participants (students, nutritionists and health workers) will give their written informed consent before they are interviewed, surveyed or involved in user acceptance testing and they will be informed about the purpose of the research, how the data can be used, the right to withdraw at anytime without penalty, and how the data will be handled to protect the privacy of the participants; they will also be informed that their dietary data could be used in anonymised form and analysed in evaluating the performance of the model, and that anonymised data could be retained for future extensions of the research. Students' participation in all research activities will be voluntary, involving no academic penalties for those who do not participate, and users of the platform will be presented as a decision-support tool – not a medical diagnosis – that is meant to complement rather than supplant clinical advice, with output presented as a range of probabilistic risk estimates rather than a medical diagnosis.

Dietary data and personal information will be kept confidential: Only the participant and authorized administrators will be able to see individual dietary logs and risk assessments; Personal information will be stored in encrypted PostgreSQL databases and will be authenticated using a JWT with password hashes; Personal information will be separated from dietary data to be used for model evaluation. The system is governed by the Law N° 058/2021 on the Protection of Personal Data and Privacy, which defines health-related data as sensitive personal data, which must be collected with explicit consent, with limited purposes and by keeping data within Rwanda if possible. The likelihood of stigmatisation due to labels like 'diabetes risk' will be reduced by: reporting the scores as probabilities with confidence intervals, emphasising actionable changes to diet over labels of 'diabetes risk', and keeping individual risk information private to the user; while the risk of algorithmic bias across age, sex and rural-urban subgroups will be addressed by explicit subgroup performance evaluation with transparent reporting in the final thesis. Interview recordings, survey answers, and recordings from user testing will be held only for the time required for analysis of the

research, and will be securely deleted within 6 months of the project's completion, unless utilized in an operational pilot; the operational data in any deployed pilot will be retained according to ALU's institutional data retention policies.

3.8 Risk Mitigation Strategies

Several technical, operational, and research risks could impact project success. The following mitigation strategies address identified risks:

ML Model Fails to Achieve AUROC ≥ 0.75 Target. Mitigation: If the multi-output XGBoost model underperforms on one or more diseases, fall back to per-disease models trained independently, apply additional feature engineering (interaction terms, polynomial features), and increase the use of class balancing techniques (class weighting, SMOTE). If a specific disease remains under threshold, scope it out of Phase 1 and report the empirical finding.

Free-Tier Infrastructure Limitations. Mitigation: Render's free tier has computational and storage limits that could be exceeded. Monitor resource usage throughout development and testing. If limits are approached, optimize database queries, implement caching for frequently accessed data (food composition lookups), and compress stored files. Request institutional support for minimal paid hosting (USD 7 per month) if free tier proves insufficient during testing.

RDHS Data Access Delay. Mitigation: The DHS Program access request typically takes 1–2 weeks. Submit the access request in Week 1 and use the waiting period to compile the Rwandan Food Composition Database from FAO INFOODS and Congo Basin FCT, which require no approval. If RDHS access is delayed beyond Week 4, supplement with publicly available CFSVA 2022 aggregate data and proceed with model development on that sample.

Incomplete Rwandan Food Composition Data. Mitigation: Many traditional Rwandan dishes will not appear directly in any FCD. For these foods, decompose recipes into ingredient components and compute nutrient values as weighted sums. Document all such estimations transparently in the FCD metadata. If a critical food remains absent, exclude it from analysis with a documented note.

Imbalanced Outcome Distribution. Mitigation: Anemia, diabetes, and overweight are minority outcomes in the RDHS sample. Apply class-weighted training to penalize false negatives, use SMOTE during training to artificially balance the minority classes, and evaluate on recall (sensitivity) and AUROC rather than raw accuracy to ensure the model is not lazily defaulting to the majority class.

Low Participation in User Acceptance Testing. Mitigation: Recruit participants early in the project timeline (by Week 8). Offer flexible testing schedules (in-person and remote options). Limit testing sessions to 30–45 minutes to respect participants' time. Provide a clear value proposition explaining how their feedback will improve the final system.

Critical Usability Issues Discovered During Testing. Mitigation: Conduct informal usability testing with 2–3 users in Week 12 before formal user acceptance testing in Week 14. This allows time to address major usability problems before final evaluation. Prioritize fixes based on severity and impact on core functionality.

Development Delays Due to Technical Complexity. Mitigation: Follow Agile one-week sprints with clearly defined deliverables. Implement a Minimum Viable Product (MVP) approach, prioritizing core features (dietary logging, food-to-nutrient conversion, multi-output risk prediction) over advanced features (trend dashboard, anonymized population aggregations). If timeline pressures emerge, reduce the trend dashboard scope to essential metrics only.

Participant Data Privacy Concerns. Mitigation: Implement encryption at rest and in transit for all data. Conduct a security review of authentication and authorization logic before user testing. Limit access to production data to the project lead only. Use synthetic test data rather than real participant records wherever possible during development. Confirm in-country data residency where feasible under Rwanda's Law N° 058/2021.

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